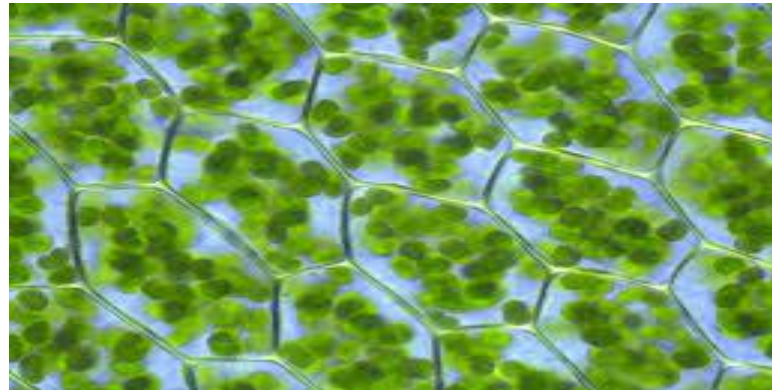
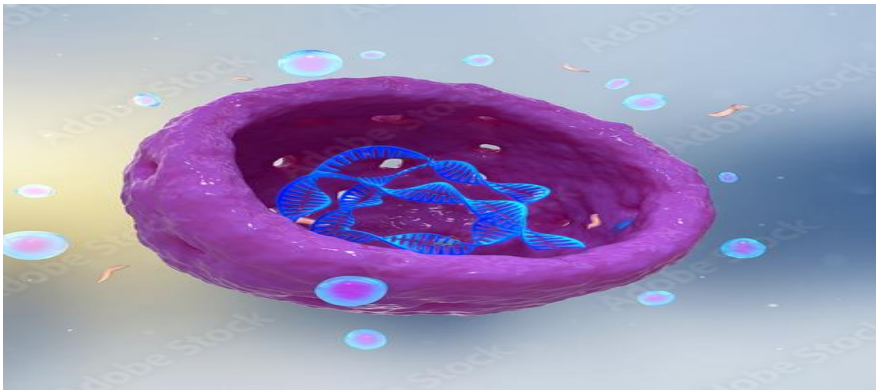




# Cell Biology

## Lecture 11



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## **Program/Course Learning Outcomes:**

### **Program Outcomes (NARS 2017):**

Demonstrate understanding of knowledge of pharmaceutical, biomedical, social, behavioral, administrative, and clinical sciences.

### **Course Learning Outcomes:**

**c1:** Interpret principles of cell structures, cell signaling, gene regulation, protein synthesis and mutation in treatment of disease.

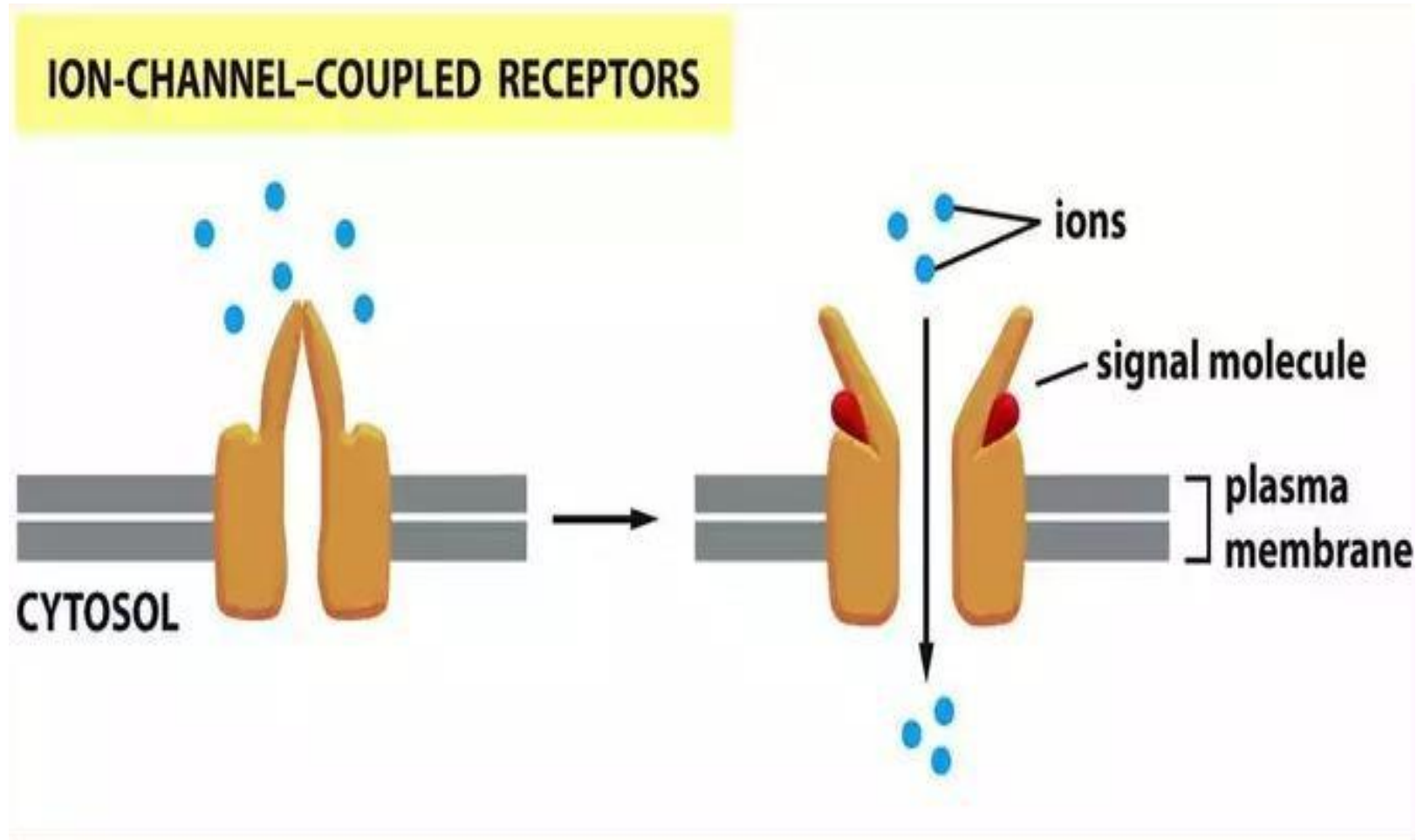
# Lecture outline

- ▶ **Cell-Surface Receptors**
- ▶ 1. Ion channel-linked receptors
- ▶ 2. Enzyme-linked receptors
- ▶ 3. G-protein linked receptors

Cell surface receptors are generally classified into the following categories:

1. Ligand-gated ion channel-linked receptors
2. Enzyme-linked receptors
3. G-protein-linked receptors

## ➤ 1. Ion channel-linked receptors



# ➤ 1. Ion channel-linked receptors

- ❖ Ionotropic receptors, responsible for regulating the transduction of chemical signals across the cell membrane in response to the chemical messenger (e.g., neurotransmitter) binding.
- ❖ This class of receptor regulates the opening or closing of ion channels that allow ions like  $\text{Na}^+$ ,  $\text{K}^+$ ,  $\text{Ca}^{2+}$ , or  $\text{Cl}^-$ , etc. to move across the plasma membrane.
- ❖ Ion channels are pore-forming proteins also referred to as cell-membrane bound receptors.

# ➤ 1. Ion channel-linked receptors

- ❖ Ions pass down their electrochemical gradient through ion channels without requiring ATP or metabolic energy.
- ❖ Ion channels are mostly found on synaptic structures which are primarily (but not exclusively) involved in neuronal activities.
- ❖ They are, therefore, an important component of the nervous system because they mediate conduction across nerve synapses when activated by neurotransmitters.
- ❖ Ion channels also play a vital role in exerting cellular response to toxins and venoms.

## ➤ 1. Ion channel-linked receptors

- ❖ Various biological processes involving fast changes in cells such as the contraction of cardiac, skeletal, and smooth muscles, activation of T-cells, and the release of hormones are also mediated through ion channels.



# ➤ 1. Ion channel-linked receptors

## ✓ Downstream mechanism

- ❖ When a ligand binds to ion-channel linked receptors, the extracellular domain of the receptor undergoes changes in its conformation, opening a channel across the plasma membrane.
- ❖ This allows specific ions (such as  $\text{Na}^+$ ,  $\text{Ca}^{2+}$ ,  $\text{H}^+$ , and  $\text{Mg}^{2+}$ , etc.) or other key molecules to pass through the open channel.
- ❖ The membrane-spanning region of these receptors helps to form a channel through which ions can pass.
- ❖ Ligands that bind to ion channel-linked receptors include neurotransmitters and peptide hormones and the passing molecules are ions such as sodium ( $\text{Na}^+$ ) and potassium ( $\text{K}^+$ ).

# ➤ 1. Ion channel-linked receptors

## ✓ Downstream mechanism

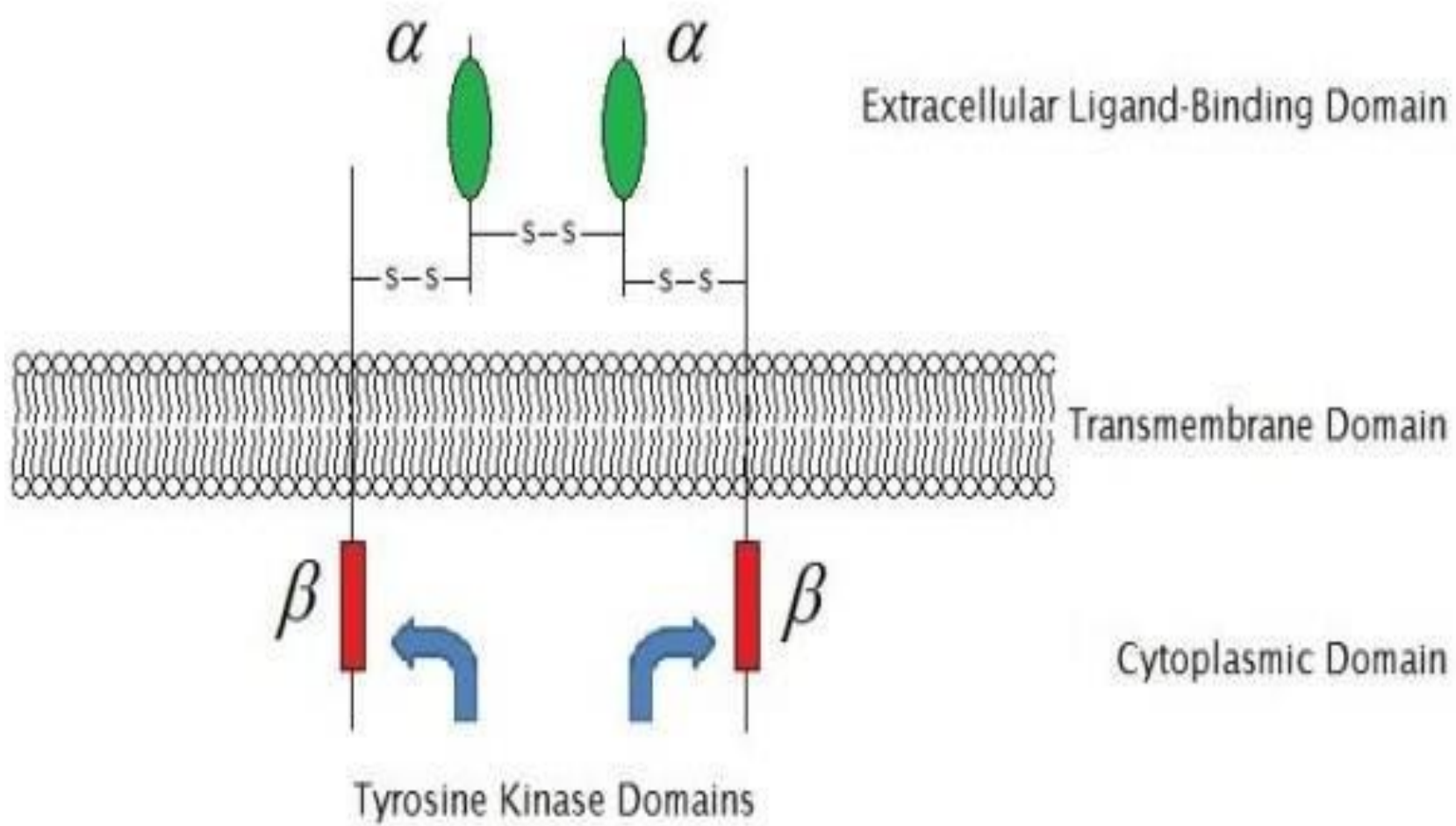
- ❖ The amino acids that occupy the membrane-spanning region of ion-channel receptors are hydrophobic in nature, making it easier for the membrane phospholipid fatty acid chains to interact with them.
- ❖ On the other hand, amino acids lining the inside of these channels are hydrophilic, allowing easy passage of ions and water.

# ➤ 1. Ion channel-linked receptors

## ✓ Downstream mechanism

- ❖ The ion channels (or pores) remain open only for a limited time, after which, the ligand dissociates from the receptor making it available to bind with a new ligand.
- ❖ Non-chemical stimuli can also cause the ion channel receptors to act in the same way. Such non-chemical stimuli include changes in electrical charge or mechanical disturbances of the membrane.

## ➤ 2. Enzyme-linked receptors



## ➤ 2. Enzyme-linked receptors

- ❖ They are typically single-pass transmembrane proteins that act as enzymes or are associated with enzymes.
- ❖ They have both an extracellular binding site for chemical signaling and an intracellular domain whose catalytic action is controlled by the binding of an extracellular ligand and are thus also called catalytic receptors.

## ➤ 2. Enzyme-linked receptors

**There are 6 types of enzyme-linked receptors:**

1. Receptor tyrosine kinases which phosphorylate specific tyrosine residues on specific intracellular signaling proteins (EGFR); they bind to polypeptide growth factors which are responsible for controlling cell proliferation and differentiation.
2. Tyrosine-kinase-associated receptors which are associated with intracellular enzymes that have tyrosine kinase activity (Cytokines).

## ➤ 2. Enzyme-linked receptors

**There are 6 types of enzyme-linked receptors:**

3. Receptor-like tyrosine phosphatases which remove phosphate groups from tyrosines of their target intracellular proteins.
4. Receptor serine/threonine kinases which phosphorylate specific serines or threonines on associated gene regulatory proteins.
5. Receptor guanylyl cyclases which directly catalyze the production of cyclic GMP in the cytosol.

## ➤ 2. Enzyme-linked receptors

**There are 6 types of enzyme-linked receptors:**

6. Histidine-kinase-associated receptors which activate a 2-component signaling pathway where the kinase phosphorylates itself on histidine residues (autophosphorylation) and then immediately transfers the phosphate to a second intracellular protein; not present in animal cells.



## ➤ 2. Enzyme-linked receptors

### ✓ Downstream mechanism

- ❖ When the enzyme-linked receptor or an enzyme associated with this type of receptor is activated, a multitude of intracellular pathways are effectively regulated.
- ❖ The intracellular domains of enzyme-linked receptors are associated with an enzyme, directly interact with an enzyme, or itself is the enzyme.
- ❖ Irrespective of large intracellular and extracellular domains of enzyme-linked receptors, a single alpha-helical region of the peptide chain is responsible for forming the membrane-spanning region of enzyme-linked receptors.

## ➤ 2. Enzyme-linked receptors

### ✓ Downstream mechanism

- ❖ Activation of the enzyme takes place as a result of the signal transmitted through the channel after a ligand binds to the extracellular domain of the receptor.
- ❖ In most cases, activation of the enzyme takes place because the receptor is dimerized upon binding of a ligand.
- ❖ The activated enzyme leads to an intracellular cascade of events executing the response.

### ➤ 3. G-protein linked receptors

- ❖ G-protein-coupled receptors (GPCRs) are the largest cell surface receptors, composed of 7 transmembrane proteins in the plasma membrane.
- ❖ GPCRs are responsible for activating the trimeric membrane-bound G-proteins (GTP binding proteins) which subsequently activate either an ion channel (effector) or an enzyme in the cell membrane.

### ➤ 3. G-protein linked receptors

- ❖ G-proteins proteins function as an intermediate transducer molecule that plays a vital role in regulating intracellular functions through a secondary mechanism which is in turn activated by G-protein coupled receptors.
- ❖ Many different types of G-protein- coupled receptors are known, such as the acetylcholine (Ach) receptor,  $\beta$ -adrenergic receptor, metabotropic glutamate receptors, certain olfactory receptors, receptors for peptide hormones, and rhodopsin (a photosensitive receptor).

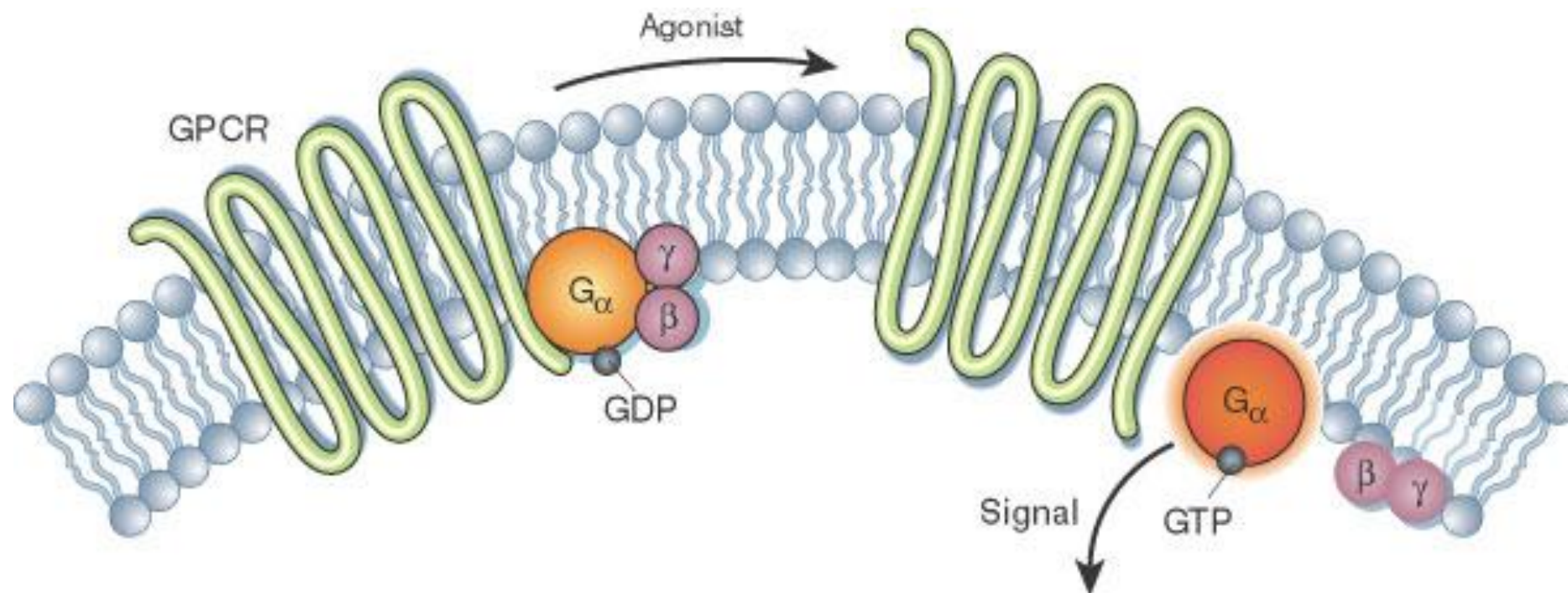
## ➤ 3. G-protein linked receptors

- ❖ Many neurotransmitters, neuropeptides, peptide hormones, as well as a number of others can activate G-protein coupled receptors.
- ❖ GPCRs are responsible for targeting various signaling pathways, including sensory perception such as sight, taste, smell, and pain sensations.
- ❖ GPCRs are among the most important cell surface receptors, with almost half of the drugs we utilize exerting their action by modifying these receptors.

## ➤ 3. G-protein linked receptors

### ✓ **Downstream mechanism**

- ❖ When the alpha subunit is reversibly bound to GDP, GPCR is inactivated.
- ❖ However, the GDP can be exchanged for GTP with the help of a guanine-nucleotide exchange factor (GEF), allowing for receptor activation.
- ❖ After a ligand binds to GPCR, the G-protein is activated. The activated G-protein, in turn, activates either an ion channel (effector) or an enzyme in the cell membrane.
- ❖ A cyclic series of events takes place as a result of cell signaling with G-protein- linked receptors.



<https://www.youtube.com/watch?v=FFvKwUSKSbc>

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